MACHINE LEARNING BASED BUYER SEGMENTATION AND INVESTMENT PROFILING FOR REAL ESTATE MARKET INTELLIGENCE

Internship Project Report

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ABSTRACT

The real estate industry generates large volumes of customer and property transaction data. Traditional customer analysis methods often fail to identify hidden behavioral patterns among buyers, resulting in inefficient marketing strategies and poor investment targeting.

This project presents a Machine Learning-based Buyer Segmentation and Investment Profiling System for Real Estate Market Intelligence. The objective is to identify distinct buyer groups using clustering techniques and provide actionable insights for business decision-making.

The project utilizes two datasets consisting of 2,000 customer records and 10,000 property transaction records. Data preprocessing techniques such as data cleaning, feature engineering, label encoding, and feature scaling were performed before applying clustering algorithms.

The K-Means clustering algorithm was used to segment buyers into five distinct groups. The Elbow Method was employed to determine the optimal number of clusters, while the Silhouette Score was used to evaluate clustering performance. The obtained Silhouette Score was 0.163, indicating meaningful customer segmentation.

The results reveal different buyer categories including Investment-Oriented Buyers, Loan-Dependent Buyers, Self-Financed Home Buyers, Corporate Investors, and International Investors. These insights can help real estate companies improve customer targeting, investment recommendations, and marketing effectiveness.

Keywords: Machine Learning, Buyer Segmentation, K-Means Clustering, Real Estate Analytics, Customer Profiling, Data Mining.

CHAPTER 1: INTRODUCTION

1.1 Background

The real estate market is one of the most dynamic sectors of the global economy. Real estate organizations interact with different types of buyers, including first-time home buyers, corporate investors, luxury investors, and international customers.

Understanding buyer behavior is essential for improving marketing campaigns, property recommendations, and investment strategies. However, traditional analysis methods often treat all customers similarly, leading to inefficient decision-making.

Machine Learning techniques provide an effective solution by identifying hidden patterns within customer data. Clustering algorithms can automatically group buyers with similar characteristics and purchasing behaviors.

This project uses Machine Learning-based clustering techniques to segment real estate buyers and generate investment profiles that support data-driven decision-making.

1.2 Problem Statement

Real estate companies often face challenges such as:

- Poor customer targeting
- Generic property recommendations
- Inefficient marketing expenditure
- Limited understanding of buyer behavior
- Difficulty identifying investment-oriented customers

Without customer segmentation, organizations cannot effectively personalize services or optimize investment strategies.

Therefore, a Machine Learning-based buyer segmentation system is required to identify meaningful customer groups and improve business intelligence.

1.3 Objectives

The main objectives of this project are:

1. Analyze buyer and property transaction datasets.
2. Perform data cleaning and preprocessing.
3. Create meaningful customer features for analysis.
4. Apply Machine Learning clustering techniques.
5. Identify hidden customer segments.
6. Generate investment profiles for each buyer group.
7. Support data-driven marketing and investment decisions.

1.4 Scope of the Project

The scope of the project includes:

- Customer behavior analysis
- Buyer segmentation
- Investment profiling
- Real estate market intelligence
- Data visualization
- Business decision support

The project focuses on unsupervised Machine Learning techniques and does not involve prediction or classification tasks.

1.5 Advantages of the Proposed System

The proposed system provides several benefits:

- Improved customer understanding
- Better marketing strategies

- Personalized property recommendations
- Efficient resource allocation
- Enhanced investment intelligence
- Identification of high-value customer segments

CHAPTER 2: DATASET DESCRIPTION

2.1 Client Dataset

The client dataset contains information about real estate buyers.

Dataset Size

Total Records: 2,000

Total Attributes: 12

Features

Feature	Description
client_id	Unique customer identifier
client_type	Individual or Company
first_name	Customer first name
last_name	Customer last name
date_of_birth	Customer date of birth
gender	Gender
country	Country of residence
region	Geographic region
acquisition_purpose	Home or Investment
satisfaction_score	Customer satisfaction rating
loan_applied	Loan application status
referral_channel	Source of customer acquisition

2.2 Property Dataset

The property dataset contains transaction information.

Dataset Size

Total Records: 10,000

Total Attributes: 9

Features

Feature	Description
listing_id	Property identifier
tower_number	Tower number
transaction_date	Transaction date
unit_category	Property type
unit_number	Unit number
floor_area_sqft	Property size
sale_price	Property price
listing_status	Sold or Available
client_ref	Linked customer identifier

CHAPTER 3: DATA PREPROCESSING

3.1 Data Quality Analysis

The datasets were analyzed for missing values and duplicate records.

Client Dataset Results

Missing Values: 0

Duplicate Records: 0

Property Dataset Results

Missing Values in client_ref: 2,695

Duplicate Records: 0

The missing values corresponded to properties with listing status marked as Available. Since these properties had not been sold, customer references were naturally unavailable.

Therefore, these records were retained.

3.2 Feature Engineering

An Age feature was generated from the date of birth field.

Age Calculation:

$$\text{Age} = \text{Current Year} - \text{Birth Year}$$

This feature provides valuable information for customer behavior analysis.

3.3 Data Encoding

Machine Learning algorithms require numerical data.

The following categorical attributes were converted into numerical values using Label Encoding:

- client_type
- gender

- country
- region
- acquisition_purpose
- loan_applied
- referral_channel

3.4 Feature Scaling

Since features exist on different scales, StandardScaler was used to normalize all variables.

Benefits:

- Improved clustering accuracy
- Faster convergence
- Balanced feature contribution

CHAPTER 4: METHODOLOGY

The overall workflow of the project is shown below:

Data Collection

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Data Cleaning

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Feature Engineering

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Data Encoding

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Feature Scaling

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K-Means Clustering

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Elbow Method

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Silhouette Score Evaluation

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Buyer Profiling

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Business Insights

The methodology ensures that customer data is transformed into meaningful segments for business intelligence and market analysis.

CHAPTER 5: EXPLORATORY DATA ANALYSIS (EDA)

5.1 Introduction

Exploratory Data Analysis (EDA) was performed to understand customer behavior, identify trends, and discover meaningful insights before applying Machine Learning algorithms.

The analysis focused on buyer demographics, acquisition purposes, financing behavior, and customer satisfaction levels.

5.2 Gender Analysis

Results

Gender	Count
Male	1012
Female	988

Interpretation

The dataset contains a nearly equal distribution of male and female buyers. Male buyers account for approximately 50.6% of the customer base, while female buyers account for 49.4%.

This balanced distribution helps eliminate gender bias during clustering and customer segmentation.

5.3 Client Type Analysis

Results

Client Type	Count
Individual	1897
Company	103

Interpretation

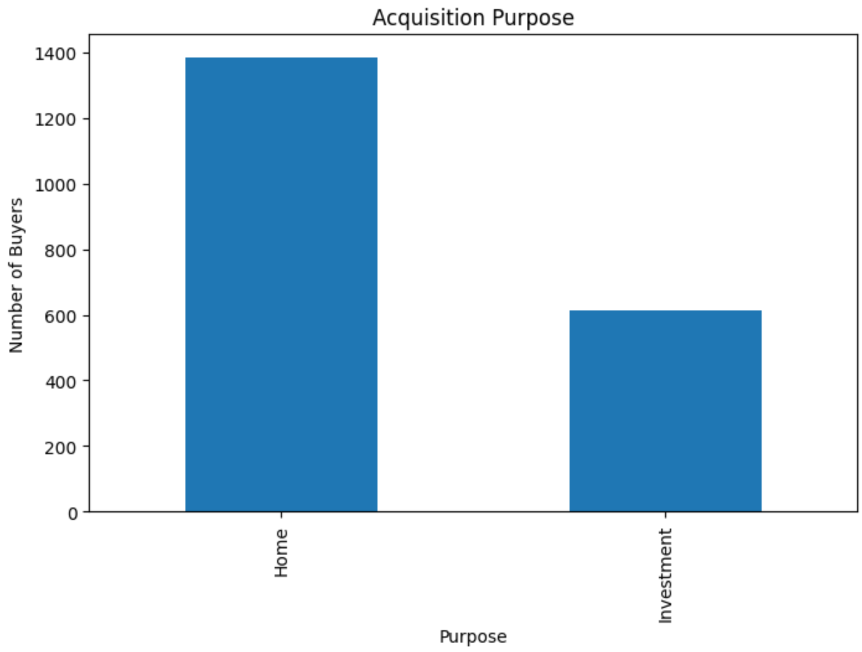
Individual customers dominate the real estate market and represent nearly 95% of all buyers.

Corporate buyers account for only 5% of the dataset, indicating that most transactions are driven by personal property purchases.

5.4 Acquisition Purpose Analysis

Results

Purpose	**Count**
Home	1385
Investment	615



Interpretation

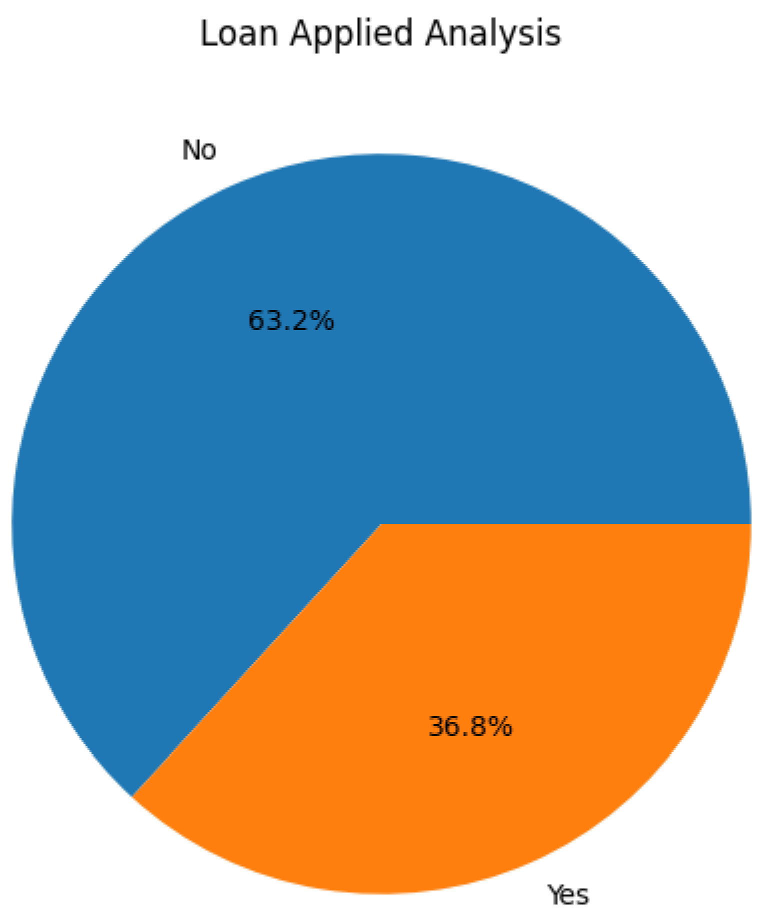
Approximately 69% of buyers purchased properties for residential use, while 31% purchased properties for investment purposes.

This indicates that home ownership remains the primary reason for property acquisition.

5.5 Loan Analysis

Results

Loan Applied	Count
No	1264
Yes	736



Interpretation

Most buyers acquired properties without requiring loans.

Around 63% of customers purchased properties independently, while 37% relied on financing options.

This finding suggests a strong presence of financially capable buyers within the market.

5.6 Country Analysis

The majority of customers belong to the United States.

Top Countries

Country	Buyers
USA	1538
UK	95
Canada	85
Germany	56
France	53

Interpretation

The United States contributes over 75% of all buyers, making it the dominant customer region in the dataset.

5.7 Customer Satisfaction Analysis

Statistical Summary

Mean Satisfaction Score = 3.03

Minimum Score = 1

Maximum Score = 5

Interpretation

Customer satisfaction levels are moderate, with an average score of approximately 3 out of 5.

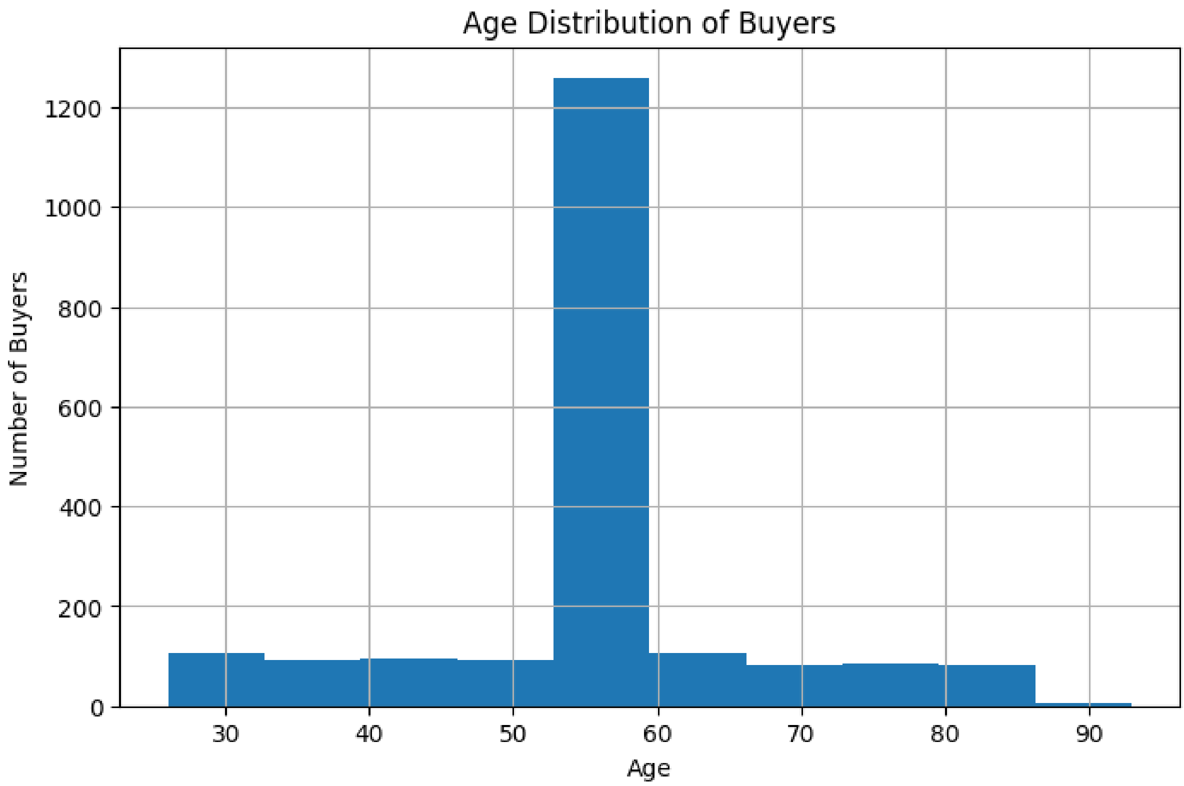
The variation in scores suggests that customer experiences differ significantly across buyer groups.

5.8 Age Distribution Analysis

The age distribution analysis revealed that most buyers belong to middle-aged and senior age groups.

The highest concentration of customers falls between 50 and 65 years of age.

This indicates that financially established individuals dominate property purchases.



CHAPTER 6: MACHINE LEARNING IMPLEMENTATION

6.1 Clustering Approach

Since the project objective was to identify hidden customer groups without predefined labels, an unsupervised learning approach was selected.

K-Means Clustering was chosen due to:

- Simplicity
- Scalability
- Interpretability
- Industry adoption

6.2 Elbow Method

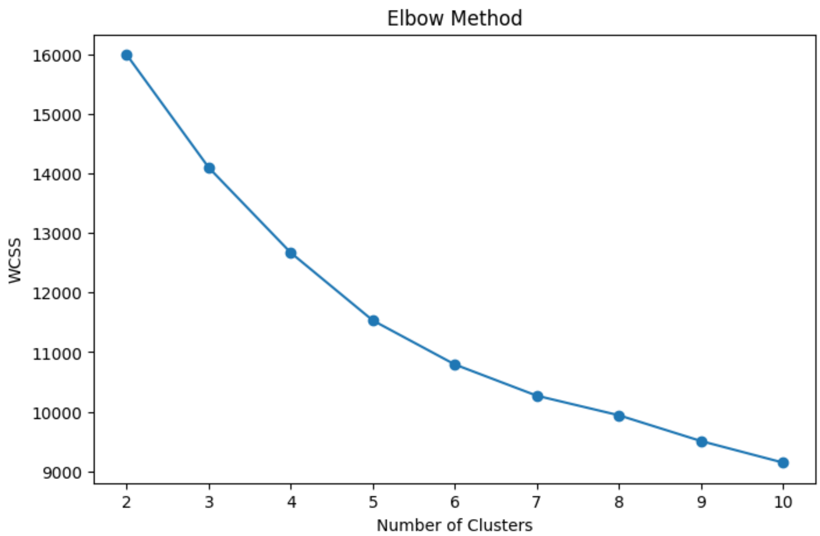
The Elbow Method was used to determine the optimal number of clusters.

WCSS (Within Cluster Sum of Squares) values were calculated for cluster sizes ranging from 2 to 10.

The resulting graph showed a clear bend at:

K = 5

Therefore, five clusters were selected for segmentation.

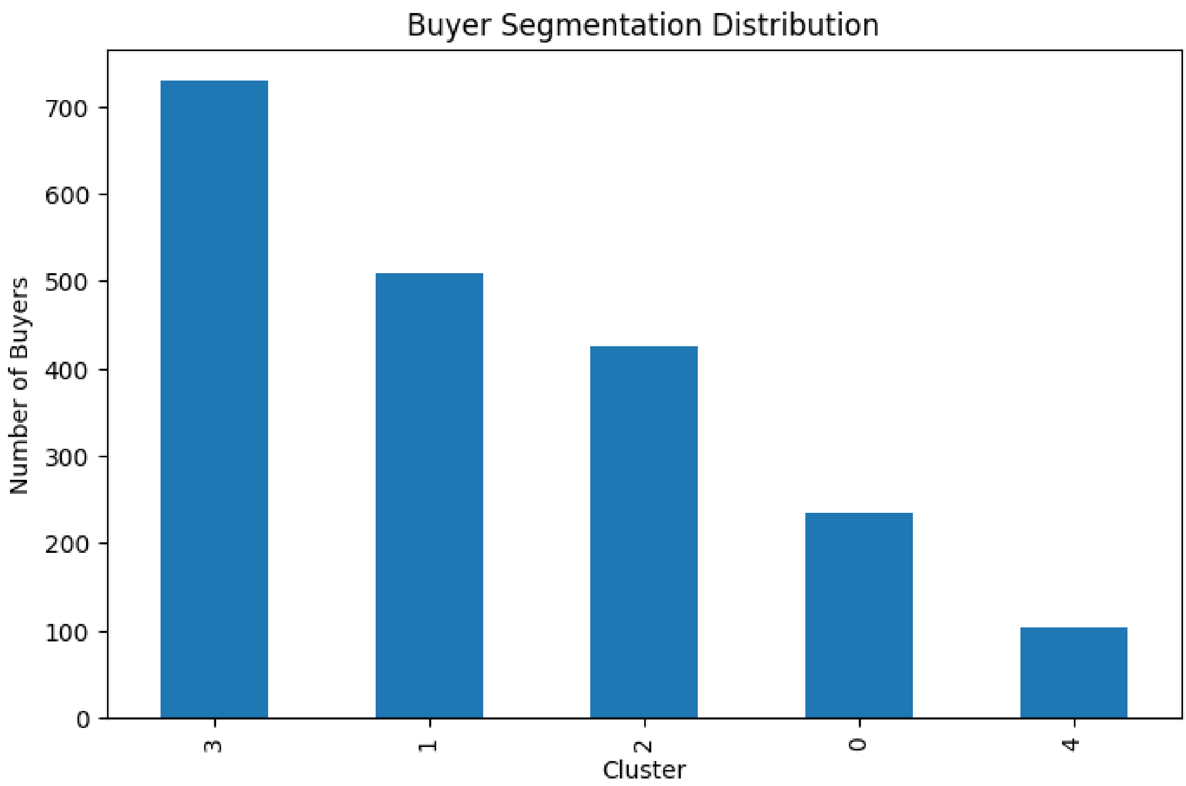


6.3 K-Means Clustering

The K-Means algorithm grouped customers based on:

- Client Type
- Gender
- Country
- Region
- Acquisition Purpose
- Loan Behavior
- Referral Channel
- Satisfaction Score
- Age

The algorithm successfully generated five customer clusters.



6.4 Cluster Distribution

Cluster	Customers
Cluster 0	234
Cluster 1	508
Cluster 2	426
Cluster 3	729
Cluster 4	103

Interpretation

Cluster 3 is the largest segment, while Cluster 4 is the smallest.

This demonstrates the existence of multiple customer categories with different purchasing behaviors.

6.5 Silhouette Score Evaluation

Result

Silhouette Score = 0.163

Silhouette Score

0.163

Interpretation

The obtained score indicates meaningful segmentation despite overlapping customer characteristics.

Real estate buyer behavior is complex and influenced by multiple factors, making moderate separation expected.

The clustering model successfully identified actionable customer groups suitable for business intelligence applications.

CHAPTER 7: BUYER SEGMENTATION RESULTS

Cluster 0: International Investors

Characteristics:

- Investment-focused buyers
- Moderate loan dependency
- Average age approximately 54 years

Business Strategy:

- Promote premium investment opportunities
- Offer international investment support

Cluster 1: Investment-Oriented Buyers

Characteristics:

- 100% investment purpose
- Moderate financing behavior
- High asset acquisition interest

Business Strategy:

- Promote rental income properties
- Offer portfolio diversification opportunities

Cluster 2: Loan-Dependent Home Buyers

Characteristics:

- 100% loan usage
- Residential property preference
- Average age approximately 56 years

Business Strategy:

- Provide financing assistance
- Partner with mortgage institutions

Cluster 3: Self-Financed Home Buyers

Characteristics:

- No loan dependency
- Largest buyer segment
- Home ownership focused

Business Strategy:

- Promote premium residential properties
- Offer loyalty and referral programs

Cluster 4: Corporate Investors

Characteristics:

- Company-based buyers
- Investment-oriented purchases
- Smaller but valuable segment

Business Strategy:

- Offer commercial investment opportunities
- Develop enterprise partnership programs

CHAPTER 8: RESULTS AND DISCUSSION

The project successfully achieved its objectives by identifying five distinct buyer groups.

Major findings include:

- Home buyers dominate the market.
- Most customers purchase properties without loans.
- The United States is the dominant customer region.
- Corporate investors form a unique customer segment.
- Customer satisfaction remains moderate across the market.

The clustering model provides valuable insights for targeted marketing, investment planning, and customer relationship management.

CHAPTER 9: CONCLUSION

This project developed a Machine Learning-based Buyer Segmentation and Investment Profiling System for Real Estate Market Intelligence.

Using customer and property transaction datasets, meaningful buyer segments were identified through K-Means Clustering.

The Elbow Method determined the optimal cluster count as five, while the Silhouette Score validated clustering effectiveness.

The generated customer profiles can support:

- Marketing optimization
- Customer targeting
- Investment planning
- Property recommendation systems
- Business intelligence initiatives

The project demonstrates the practical application of Machine Learning in real estate analytics.

CHAPTER 10: FUTURE SCOPE

Future improvements may include:

1. Advanced clustering algorithms such as DBSCAN and Hierarchical Clustering.
2. Predictive analytics for customer purchase behavior.
3. Real-time dashboard implementation using Streamlit.
4. Property recommendation systems.
5. Integration with cloud-based analytics platforms.
6. Deep Learning-based customer profiling.

These enhancements can further improve customer intelligence and decision-making capabilities.

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APPENDIX

Figure A1: Sample Records from Client and Property Datasets

	listing_id	tower_number	transaction_date	unit_category	unit_number	floor_area_sqft	sale_price	listing_status	client_ref
0	1012	1	01-01-2024	Apartment	12	1160.36	$300,385.62	Sold	C0027
1	1015	1	01-01-2024	Apartment	15	782.25	$208,930.81	Sold	C0097
2	1021	1	01-01-2024	Apartment	21	756.21	$218,585.92	Sold	C0113
3	1030	1	01-01-2024	Apartment	30	743.09	$246,172.68	Sold	C0141
4	2016	2	01-01-2024	Apartment	16	701.66	$212,265.67	Sold	C0146

Figure A2: Missing Value Analysis

The client dataset contains no missing values. The property dataset contains 2,695 missing values in the client_ref column, which correspond to available properties that have not yet been sold.

	0
listing_id	0
tower_number	0
transaction_date	0
unit_category	0
unit_number	0
floor_area_sqft	0
sale_price	0
listing_status	0
client_ref	2695

dtype: int64

Figure A3: Cluster Distribution Counts

The clustering model segmented customers into five groups, with Cluster 3 being the largest segment and Cluster 4 being the smallest.

```
Cluster
3    729
1    508
2    426
0    234
4    103
Name: count, dtype: int64
```